BIBLIOMETRIC NETWORK STUDY · 2014–2025

Using Bipartite Network Analysis to Examine Global Collaboration between Engineering and STEM Education

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INTRODUCTION

Two fields growing fast — but apart

- Engineering education and STEM education have both grown rapidly, yet are usually studied in isolation.
- Co-authorship reveals how knowledge is produced — and where the two fields connect.
- Cross-domain links between them remain underexplored.

Research Questions

1. What does the collaboration network look like?
2. How vital are cross-domain authors?
3. Do country and field shape who collaborates?

Collecting the data & building the network

1 · Data Collection

- **Source: Scopus, 2014–2025**
- 10 journals — 5 engineering education + 5 STEM education
- Extracted: authors, Scopus IDs, year, source, affiliation, EID
- **Final corpus: 6,407 articles · 13,212 unique authors**

2 · Build the Network

- Construct an author–paper bipartite network (authors linked to the papers they wrote).
- Classify each author by publication history:
- *Engineering-only · STEM-only · Boundary-spanning (≥1 paper in each field)*
- Project into a one-mode co-authorship network (author ↔ author).

What we measured — and what it means

Density

How many of all possible co-author ties actually exist. Low density = a sparse network.

Giant component

The largest set of authors all reachable through chains of co-authorship — a proxy for how connected the field is.

Clustering

How tightly authors form closed collaboration groups (cliques).

Node-removal test

Delete the boundary-spanning authors, then re-measure the giant component to see how much they hold the network together.

ERGM

A statistical model of tie formation. Tests whether authors prefer collaborators from the same country (country homophily) or same field (field homophily).

RESULT 1

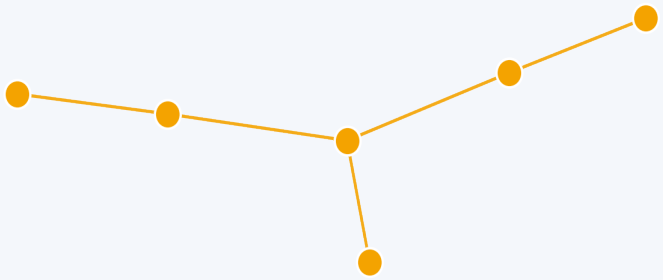
Answers RQ 1

Sparse but connected — STEM is more integrated

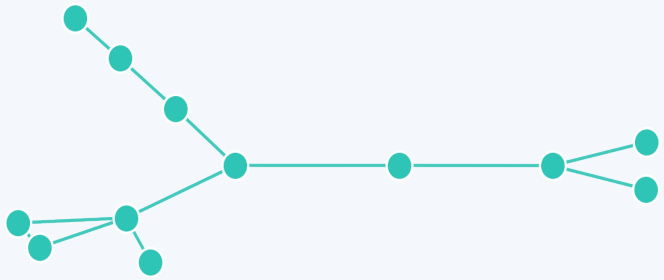
How integrated is each field? STEM forms a larger connected core.

Low density, large reach

Engineering education



STEM education



Sparse, yet connected



giant component = 29% of authors

in the giant component · detached / fragmented

giant component = 54% of authors

few direct ties — yet ~47% link into one component

reachable via chains (giant component) · outside the giant component

46.4%

of nodes in the giant component (9,100 of 19,619)

0.78

clustering in both subdomains — tight-knit groups

Takeaway

The network is **sparse but connected** — most authors never co-author directly, yet chains link nearly half into one component. STEM education forms a markedly larger connected core than engineering education.

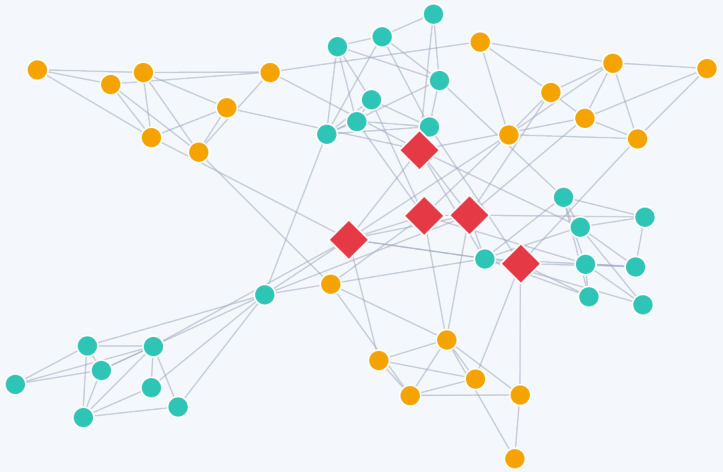
RESULT 2

Answers RQ 2

3.4% of authors hold the network together

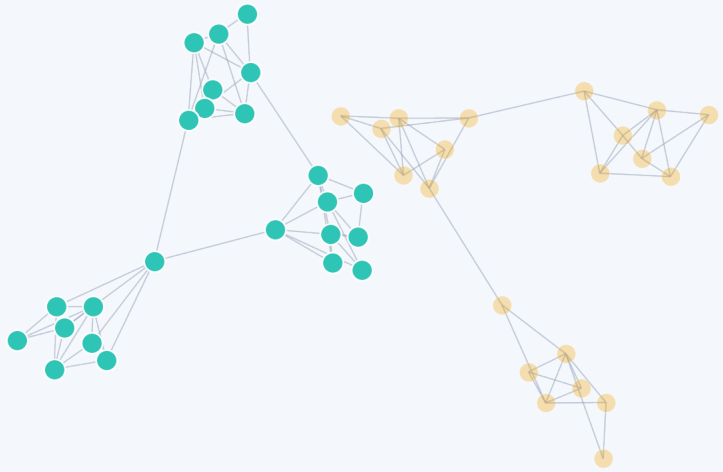
Removing the 3.4% of cross-domain authors collapses the giant component by ~54%

Before · full network



Giant component intact

After · cross-domain authors removed



Network fragments into isolated clusters

Engineering education author · STEM education author · Cross-domain (bridge) author

Stylized illustration — not actual network topology

451
cross-domain authors
3.4% of 13,212

−54%
giant component collapse
9,100 → 4,189 nodes

1.9×
higher average degree
vs. engineering-only

Geography is concentrated: the U.S. holds 374 of 451 cross-domain authors. Migration tilts engineering → STEM (49.9% vs. 38.1%).

RESULT 3

Answers RQ 2

Cross-domain collaboration rose, then declined

Share of papers involving cross-domain authors, 2014–2025

Reading the trend

2014–2016: stable at 15–18%

2017–2021: climbs to a 25.1% peak

After 2022: declines to 19% by 2025

Integration is recent and fragile — gains can reverse.

RESULT 4

Answers RQ 3

Country matters most; field silos persist

ERGM — what makes a collaboration tie more likely?

Country homophily

Strong & significant ($p < .001$). Authors strongly prefer collaborators from the same country.

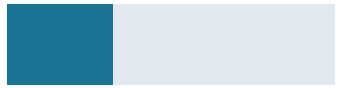
+0.118



Field homophily

Significant but weaker. Disciplinary silos between the two fields persist.

+0.038



What it means

- Geography is a bigger barrier than discipline.
- Crossing borders is harder than crossing fields.
- Both forces pull authors inward, limiting global integration.

CONCLUSIONS & RECOMMENDATIONS

Dense within, sparse between

What we found

- Strong internal clustering, but fragile cross-domain links.
- A few boundary-spanning authors keep the network whole.
- Geography is the biggest barrier — the network is easily disrupted.

What to do

Diversify the bridges

Grow the pool of boundary-spanning authors so integration rests on many, not few.

Fund across borders

Back Global North–South and East–West partnerships to offset country homophily.

Build durable mechanisms

Reward sustained cross-domain and cross-national collaboration, not one-off ties.

Thank you

Questions & discussion welcome

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